

EVAPORATIVE COOLING OF FLARE PLASMA

S. K. ANTIOCHOS AND P. A. STURROCK

Institute for Plasma Research and Department of Applied Physics, Stanford University

Received 1977 January 3; accepted 1977 September 26

ABSTRACT

We investigate a one-dimensional loop model for the evaporative cooling of the coronal flare plasma. The important assumptions are that conductive losses dominate radiative cooling and that the evaporative velocities are small compared with the sound speed. We calculate the profile and evolution of the temperature and verify the accuracy of our assumptions for plasma parameters typical of flare regions. The model is in agreement with soft X-ray observations on the evolution of flare temperatures and emission measures. The effect of evaporation is to greatly reduce the conductive heat flux into the chromosphere and to enhance the EUV emission from the coronal flare plasma.

Subject headings: plasmas — Sun: corona — Sun: flares — Sun: X-rays

I. INTRODUCTION

It is becoming widely accepted that flares evaporate large masses of chromospheric material into the corona. The most compelling arguments for the occurrence of this process come from soft X-ray observations. Determinations of the emission measure, which appears to be well over 10^{50} cm^{-3} for some flares (Phillips and Neupert 1973), indicate a total mass of hot ($T \geq 10^7 \text{ K}$) plasma in the corona as large as 10^{16} g . Similar mass estimates are obtained from loop prominence systems which are known to occur in postflare regions (Jefferies and Orrall 1965). This amount of plasma is too large to condense from the corona and must somehow be extracted from the chromosphere.

Neupert (1968) was perhaps the first to realize that coronal flare plasma is the result of evaporation from the chromosphere. Similar ideas were expressed subsequently by Hudson and Ohki (1972) and by Sturrock (1973). In these papers, it was implicitly assumed that evaporation occurs during the initial deposition of energy into the chromosphere by means of high-energy electron streams. However, there appears to be observational evidence for evaporation during the decay phase of flares. The emission measure is often seen to increase, while the total thermal energy does not increase and the temperature decreases (Kahler, Meekins, and Kreplin 1970; Horan 1971). "Moving boundary" types of models have been proposed, to explain these results, by Strauss and Papagiannis (1971) and by Zaumen and Acton (1974). These models involve the assumption that flare heating occurs only at the top of a loop of coronal plasma, which results in an expanding hot region. The loop is assumed to be essentially infinite in length so that the effect of the heating on the chromosphere is not included. However, these models encounter certain difficulties: In particular, they cannot account for the origin of the large masses required by the soft X-ray observations.

Furthermore, if the "moving boundary" models were correct, flares should have heated the corona well before they heated the chromosphere, whereas observations indicate that the $H\alpha$ emission of flares begins well before the soft X-ray maximum. We therefore take the view that the increase of emission measure during the decay phase of a flare indicates that evaporation occurs not only during the impulsive phase of a flare, but also during the decay phase.

From the above considerations, it appears that there may be two evaporation mechanisms occurring during a solar flare. One is associated with the direct heating of the chromosphere by the energy released during the impulsive phase of a flare, the energy's probably being transferred by an electron stream. This stage is likely to involve large pressure gradients and large velocities, quite possibly leading to shock waves. It seems likely that a separate evaporation process may occur after the primary energy release is finished, this evaporation's being driven by the large conductive heat flux from the high-temperature flare plasma contained in magnetic flux tubes above the chromosphere. In this stage, evaporation is a cooling process competing with radiation. It seems likely that energy loss by evaporation dominates early in the decay phase and that energy loss by radiation dominates later, when the density of flare plasma is higher and the temperature is lower. Moreover, it seems likely that this evaporation stage would involve only small pressure gradients and velocities. It is this later, "gentle" evaporation process which we discuss in this paper.

II. MODEL

Since the magnetic field strength in typical post-flare regions appears to be large ($B \geq 10$ gauss [Rust and Bar 1973]), we assume that it dominates the plasma so that mass flow and heat transfer are both parallel to the magnetic field. The geometry which we adopt

is that of a flare loop consisting of a narrow current-free flux tube, with the independent variable s measuring distance along the loop (parallel to $\mathbf{B}$) from the top of the loop. The cross-sectional area of the flux tube is given by $A(s)$ and will depend on the form of the magnetic field. The field is assumed to be that of a point dipole or of a line dipole, but we also calculate the case of a loop of constant cross section corresponding approximately to the familiar plane-parallel model.

We make two assumptions on the state of the plasma. The first is that radiation losses are negligible. We later check our assumption by comparing the radiative cooling time with the conductive cooling time to be derived below.

Our second assumption is that the velocities generated by the evaporation process are small compared with the speed of sound. Since the temperature is high enough that gravitational effects are negligible (i.e., the scale height is large compared with the size of the flux tube), this assumption is equivalent to setting

$$\frac{\partial p}{\partial s} \approx 0. \quad (2.1)$$

At this point it may seem paradoxical that we can calculate the plasma velocities by neglecting the pressure gradients, since these gradients create the velocities in the first place. However, one finds that, if the gravitational term is negligible, the fractional variation of pressure along the flux tube is of order M^2 , where the Mach number M is given by

$$M = v/C, \quad (2.2)$$

where C is the speed of sound. Hence our assumption (2.1) is equivalent to the assumption that $M \ll 1$, which must be verified *a posteriori*.

Equation (2.1) shows that the pressure is independent of s . We assume that there is no heat input into the loop, and we ignore radiation. Hence the total energy in the loop is constant, which implies that the pressure is also independent of time:

$$p(s, t) = \text{const}. \quad (2.3)$$

On using equation (2.3), one finds that the energy equation may be expressed as

$$-\frac{5}{2} \frac{p}{\rho} \left(\frac{\partial \rho}{\partial t} + v \frac{\partial \rho}{\partial s} \right) = \frac{1}{A} \frac{\partial}{\partial s} \left(A \kappa \frac{\partial T}{\partial s} \right). \quad (2.4)$$

On combining equation (2.4) with the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{1}{A} \frac{\partial}{\partial s} (A \rho v) = 0, \quad (2.5)$$

we obtain

$$\frac{\partial}{\partial s} \left[A \left(\frac{5}{2} p v - \kappa \frac{\partial T}{\partial s} \right) \right] = 0. \quad (2.6)$$

We see from equation (2.6) that

$$A \left(\frac{5}{2} p v - \kappa \frac{\partial T}{\partial s} \right) = f(t). \quad (2.7)$$

The function $f(t)$ represents the flux of enthalpy along the flux tube (from one footpoint to the other). This flux will be determined in part by conditions at the footpoints and in part by the action of the flare. The first contribution may, for instance, be significant, if one footpoint is in a sunspot umbra and the other is in a plage. It seems likely that the chromospheric Evershed effect is the visible manifestation of this process. Even in this case, we expect the mass and energy flux to be small compared with that due to a flare. If the chromospheric conditions are similar at both footpoints, there is no reason to expect an enthalpy flux along the tube. Since we have chosen to consider a *symmetrical* flux tube, there can be no nonzero enthalpy flux. We therefore set

$$f(t) = 0. \quad (2.8)$$

Using equations (2.7) and (2.8), we can relate the heat flux F , given by

$$F = -\kappa \frac{\partial T}{\partial s}, \quad (2.9)$$

to the velocity:

$$F = -\frac{5}{2} p v. \quad (2.10)$$

From this equation we note that, at the temperature maximum in the loop (which occurs at the top, if symmetry is assumed), the velocity and the heat flux are both zero.

The heat equation may now be expressed with only temperature as a dependent variable. On replacing v in equation (2.4) by using the expressions (2.9) and (2.10), and then eliminating ρ in favor of T by means of the equation of state, we obtain

$$\frac{5}{2} \frac{p}{T} \frac{\partial T}{\partial t} = \frac{1}{A} \frac{\partial}{\partial s} \left(A \kappa \frac{\partial T}{\partial s} \right) - \kappa \left(\frac{\partial T}{\partial s} \right)^2. \quad (2.11)$$

We use the Spitzer (1962) formula for thermal conductivity,

$$\kappa = \alpha T^{5/2}, \quad (2.12)$$

where, with sufficient accuracy for our purposes, we may take $\alpha = 10^{-6}$.

Equation (2.11) is nonlinear, but it is amenable to solution by separation of variables. We assume that T is of the form

$$T(s, t) = T_M \theta(t) \phi(s). \quad (2.13)$$

For convenience, we choose

$$\theta(0) = 1, \quad \phi(0) = 1, \quad (2.14)$$

so that T_M is the maximum temperature of the plasma (the initial temperature at the top of the loop). We

may substitute equation (2.13) into equation (2.11) and obtain

$$-\frac{5}{2} \frac{p}{\alpha T_M^{7/2}} \frac{d}{dt} (\theta^{-7/2}) = \frac{1}{A} \frac{d}{ds} \left(A \frac{d\psi}{ds} \right) - \frac{2}{7} \frac{1}{\psi} \left(\frac{d\psi}{ds} \right)^2, \quad (2.15)$$

where $\psi(s)$ is related to $\phi(s)$ by

$$\psi = \phi^{7/2}, \quad (2.16)$$

In equation (2.15) the variables are separated so that we may set each side equal to $-k^2$ (independent of both t and s). The equation for $\theta(t)$,

$$\frac{d}{dt} (\theta^{-7/2}) = \frac{2}{5} \frac{\alpha T_M^{7/2}}{p} k^2, \quad (2.17)$$

may be integrated to give

$$\theta(t) = (1 + t/\tau_c)^{-2/7}, \quad (2.18)$$

where

$$\tau_c = \frac{5p}{2\alpha T_M^{7/2} k^2}. \quad (2.19)$$

The equation for ψ becomes

$$\frac{d^2\psi}{ds^2} - \frac{2}{7} \frac{1}{\psi} \left(\frac{d\psi}{ds} \right)^2 + \frac{1}{A} \frac{dA}{ds} \frac{d\psi}{ds} = -k^2. \quad (2.20)$$

We see from equation (2.14) and the symmetry of the flare loop that we require that

$$\psi = 1, \quad \frac{d\psi}{ds} = 0 \quad \text{at } s = 0. \quad (2.21)$$

Given k and using the boundary conditions (2.21), we can integrate equation (2.20) to yield $\psi(s)$. It will be found that ψ vanishes at two points $s = \pm s_b$. Near those points the temperature drops very rapidly, as the gradient, in fact, is infinite at $s = \pm s_b$. It is clear that the model must break down in the neighborhood of these points. This is to be expected, since radiation becomes increasingly important as the temperature approaches a few times 10^4 K.

These considerations indicate that, from physical considerations, the base of the model should be taken to be $s = \pm s_b'$, where s_b' is the lowest height at which radiation may be neglected. However, the difference between s_b and s_b' is very small compared with s_b . For this reason, we adopt the mathematical definition of the base of our model:

$$\psi(s_b) = 0. \quad (2.22)$$

Once the corresponding model is found, the height of the "physical" base may be determined if required.

For given s_b , conditions (2.21) and (2.22) can be met only for a unique value of k in equation (2.20). This value will, of course, depend implicitly also on the area function $A(s)$. For either type of magnetic field tube (that produced by a point dipole or that

produced by a line dipole), we shall wish to consider a range of values of s_b . Rather than prescribe s_b and then determine k as an eigenvalue of the differential equation (2.20), it is clearly simpler to consider a range of values of k and determine s_b for each value. This may be done by finding a solution of the equation with the initial conditions (2.21) and then finding the point s_b at which $\psi = 0$.

As we noted above, the position of the physical base of our model (the lowest height at which radiation may be neglected) $s = \pm s_b'$ will not exactly coincide with the "mathematical" base $s = \pm s_b$. To explore this point further, we express the total radiative energy-loss rate S_R as

$$S_R = \int dT \frac{Q(T)}{T}, \quad (2.23)$$

so that $Q(T)$ is the loss rate "per unit logarithmic temperature interval." On comparing this expression with the alternative expression

$$S_R = \int ds n^2(s) \Lambda[T(s)] A(s), \quad (2.24)$$

where $n^2 \Lambda(T)$ is the volume emissivity, we see that

$$Q(T) = 2H(T)n^2 \Lambda A, \quad (2.25)$$

where n and A are to be evaluated at the locations where the temperature has the value T . H is the temperature scale height

$$H = T/|\partial T/\partial s| \quad (= n/|\partial n/\partial s|), \quad (2.26)$$

which, since $\partial p/\partial s = 0$, is also the density scale height. The factor 2 occurs because there are two points at which the temperature has the value T , one in each "leg."

A reasonable criterion that radiation energy loss is negligible compared with energy loss by heat flux is that

$$\frac{Q \Delta T/T}{2FA} \ll \frac{\Delta T}{T}, \quad (2.27)$$

where A is the area and F the magnitude of the heat flux at the two locations at which the temperature is T . This may be rewritten as the requirement that $R(T)$ be small compared with unity, where

$$R(T) = \frac{Q}{2FA} = \frac{H(T)n^2 \Lambda}{Fa}. \quad (2.28)$$

On using equations (2.9), (2.25), and (2.26), expressed in terms of temperature, we find that

$$R(T) = \frac{\alpha p^2}{4k_B^2} \frac{T^{3/2} \Lambda(T)}{F^2 a}, \quad (2.29)$$

where k_B is Boltzmann's constant.

We shall find in the next section that, for any model obtained by solving equation (2.20), the heat flux and temperature vary along the tube in such a way that

F/T is almost constant. This facilitates examination of the expression (2.29): We find that R approaches unity when the temperature approaches 10^5 K. This implies that the "physical" base should be located at a temperature level above 10^5 K. However, the temperature in this region is dropping so rapidly that the difference between this position and the "mathematical" base where $T = 0$ is negligible in comparison with the overall height of the tube.

The fact that, over most of the tube, F varies linearly with T has the important consequence that the fluxes of heat and of mechanical energy at the chromosphere are negligible in comparison with the fluxes of heat and of mechanical energy in the high-temperature part of the tube. Hence the transfer of energy between the corona and chromosphere is negligible. In our model, cooling is due principally to the evaporation (redistribution) of plasma which is already at coronal temperatures.

In this respect, the current model is in sharp contrast to the static model (Antiochos and Sturrock 1976), for which F tends to a constant value as T decreases. The corresponding difficulty, in the static model, of dissipating a large heat flux by radiation from the chromosphere has no counterpart in the present evaporative model.

III. RESULTS

a) Temperature Profile

For the case of a loop of constant cross section, equation (2.20) can be integrated directly, yielding

$$\begin{aligned} \frac{1}{2}\phi(1 - \phi^{3/2})^{1/2} - \frac{2(1 - \phi^{3/2})^{1/2}}{3^{1/2} + 1 - \phi^{1/2}} \\ - \frac{3^{1/2} - 1}{3^{1/4}} \mathcal{F}[\beta(\phi), \sin 75^\circ] \\ + 2.3^{1/4} \mathcal{E}[\beta(\phi), \sin 75^\circ] = \frac{1}{2}(7/6)^{1/2} k s, \end{aligned} \quad (3.1)$$

where

$$\beta(\phi) = \arccos \left(\frac{\sqrt{3} - 1 + \sqrt{\phi}}{\sqrt{3} + 1 - \sqrt{\phi}} \right) \quad (3.2)$$

and $\mathcal{F}$ and $\mathcal{E}$ are the elliptic integrals of the first and second kind, respectively. The constant k can now be expressed in terms of the position s_b of the base. Using equations (3.1) and (3.2), we find that, for the planar case,

$$k = 1.60/s_b. \quad (3.3)$$

For loops of variable cross section, equation (2.20) has been integrated numerically. The important parameters that characterize the geometry are the loop length s_b and the ratio of the cross-sectional area $A(0)$ at the top of the loop to the area $A(s_b)$ at the base. We term this ratio the compression factor Γ :

$$\Gamma = A(0)/A(s_b) = B(s_b)/B(0). \quad (3.4)$$

Rust and Bar (1973) find that $\Gamma \lesssim 30$ in the flare of 1972 August 7.

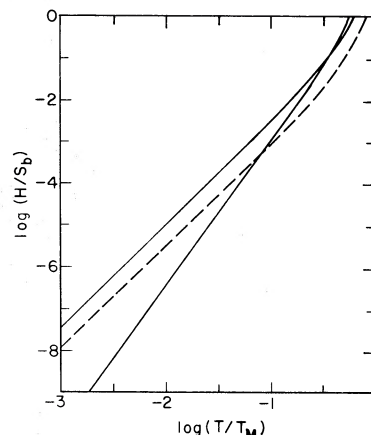


FIG. 1.—The temperature scale height as a function of temperature along the loop for three models. The top solid line refers to an evaporative model of constant cross section. The broken line refers to an evaporative model for a loop due to a line-dipole source with $\Gamma = 30$. The bottom solid line refers to a static model of constant cross section.

For a given form of $A(s)$, we may solve equation (2.20) for $\phi(s)$. In Figure 1, we plot the temperature scale height H implied by our model as a function of temperature along the loop, where H has been defined in equation (2.26). This is done for the case of a constant-area loop and for a flux tube due to a line-dipole source with a compression factor $\Gamma = 30$. The results for a point-dipole source are almost identical to those for a line-dipole source. That the scale height H becomes infinite at the top of the loop simply reflects the fact that the spatial derivative of the temperature is zero at the top of the loop. This peculiarity is without physical significance; it simply means that the spatial variation of temperature is determined by the second derivative of the temperature rather than the first derivative.

The true size scale over which T varies is, of course, less than the loop length. For $T/T_M < 10^{-1}$, the scale height is small compared with the loop length, $H(T) < 10^{-2}s_b$, and decreases as $T^{5/2}$ in both the constant-area and the variable-area case. The loop would therefore appear to an observer to be almost isothermal with a very thin transition region, as is the case for the quiet corona.

For comparison we also plot in Figure 1 the temperature scale height given by the static model discussed by Antiochos and Sturrock (1976). From this figure we find that, at low temperatures ($T < 10^{-1}$), the temperature profile is given by $T \propto s^{2/5}$ for evaporative cooling and by $T \propto s^{2/7}$ for static cooling. Hence the heat flux F is linearly proportional to T in the evaporative models, whereas $F \approx \text{const.}$ in the static models. This result argues strongly in favor of the evaporative models.

For temperatures and size scales typical of flares, the heat flux into the chromosphere implied by the static models is very large—in excess of $10^8 \text{ ergs cm}^{-2} \text{ s}^{-1}$. It is difficult for static models to account for the dissipation of such a large flux; however, in our present

model this flux is reduced by the evaporation process. The heat flux into the chromosphere is approximately three orders of magnitude lower for an evaporative model than for a static model.

b) Cooling Time Scale

We can evaluate the cooling time τ_c in equation (2.18) by using equation (3.3). For typical loop parameters,

$$T \approx 10^7 \text{ K}, \quad n \approx 10^{10.5} \text{ cm}^{-3}, \quad (3.5)$$

and

$$s_b \approx 10^{9.5} \text{ cm},$$

equations (2.19) and (3.3) yield

$$\tau_c \approx 10^2 \text{ s}. \quad (3.6)$$

The radiative cooling time is given by

$$\tau_R \approx \frac{3}{2} \frac{p}{n^2 \Lambda(T)}, \quad (3.7)$$

where $\Lambda(T)$ has been computed by Raymond, Cox, and Smith (1976). For the above parameters (3.5), we find that $\tau_R \approx 10^3 \text{ s}$; hence our assumption that conduction dominates the cooling is justified. However, as the temperature decreases, the ratio τ_c/τ_R increases, until eventually radiation dominates and evaporation becomes unimportant. For the loop described above, we expect this to occur at $T \lesssim 5 \times 10^6 \text{ K}$.

We have also calculated the cooling times for loops due to either a point-dipole or a line-dipole source. The effect of the geometry is shown in Figure 2, where we plot the ratio of the cooling times for a loop τ_c and for a planar τ_p (constant cross section) model as a function of Γ . As in the static case (Antiochos and Sturrock 1976), the effect of the compression factor is to increase the cooling time; however, we still expect that conduction dominates over radiation during the initial stages, if $T \gtrsim 10^7 \text{ K}$.

The other main assumption in our model is that the evaporation velocities are small compared with the speed of sound. For the planar case, the Mach number can be obtained directly from equations (2.9) and (3.1):

$$M = \frac{v}{c} = 10^{-10.6} \frac{T_M^3}{p s_b} \phi^{1/2} (1 - \phi^{3/2})^{1/2} \left(1 + \frac{t}{\tau_c}\right)^{-6/7}. \quad (3.8)$$

The maximum value of M occurs at $t = 0$ and $\phi = (2/5)^{2/3} \approx 0.5$. For the parameters in (3.5), we find that $M \lesssim 10^{-1}$, so that our assumption is valid. From the form of the time dependence of M and T , we find that $M \propto T^3$ for fixed s ; hence our approximation improves as the loop cools.

c) Radiation

Various authors have attempted to calculate the emission measure and the temperature of flare regions from soft X-ray observations (e.g., Kahler *et al.*; Horan 1971). Craig and Brown (1976) point out that

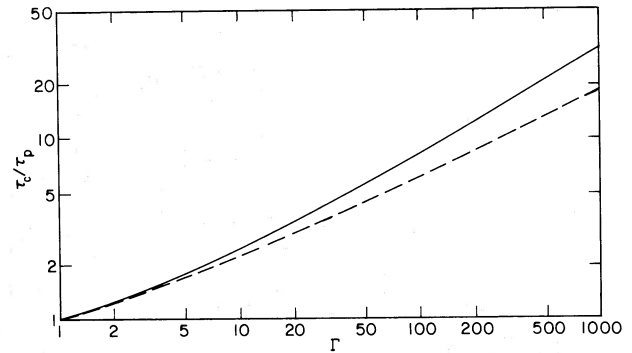


FIG. 2.—Ratio of cooling times for a loop τ_c and a planar (constant cross section) τ_p model as a function of Γ . The solid line refers to a loop due to a point-dipole source, and the broken line to a loop due to a line-dipole source.

these calculations are inherently inaccurate, and they suggest that one should instead assume a definite model for the structure of a flare region and then test this model by comparing the expected radiation with observations. We therefore calculate the evolution of the soft X-ray emission predicted by our evaporative loop model.

Generally, the observations consist of measurement of the total flux of X-rays in some energy range. Hence we must compute

$$\mathcal{L}(E_1, E_2, t) = \int_{E_1}^{E_2} dE \int_V d^3x \eta(E, x, t), \quad (3.9)$$

where the spatial integration is over the volume of the loop and η is the energy flux at the Earth from a unit volume of plasma. If the dominant radiation process is bremsstrahlung, the emissivity η ($\text{keV cm}^{-3} \text{ s}^{-1} \text{ keV}^{-1}$) is given by

$$\eta = a \exp(-E/k_B T) n^2 T^{-1/2}, \quad (3.10)$$

where $a = 10^{-38.09}$ for T in K (Kahler *et al.*). Using equations (3.1) and (3.10) and integrating equation (3.9) over E , we obtain for a loop of constant cross section A

$$\begin{aligned} \mathcal{L}(r_1, r_2, t) &= C_1 \theta^{-3/2}(t) \int_0^1 d\phi \frac{\exp(-r_1/\theta\phi) - \exp(-r_2/\theta\phi)}{(1 - \phi^{3/2})^{1/2}}, \end{aligned} \quad (3.11)$$

where

$$C_1 = 10^{-13.8} \frac{p^2 A s_b}{T_M^{3/2}} \quad (3.12)$$

and

$$r_1 = \frac{E_1}{k_B T_M}, \quad r_2 = \frac{E_2}{k_B T_M}. \quad (3.13)$$

In Figure 3 we plot $\mathcal{L}$ for the particular values $r_1 = \frac{1}{2}$ and $r_2 = 1$ as a function of t/τ_c . Assuming a maximum loop temperature $T_M = 2 \times 10^7 \text{ K}$, these values for r_1 and r_2 correspond to a soft X-ray range

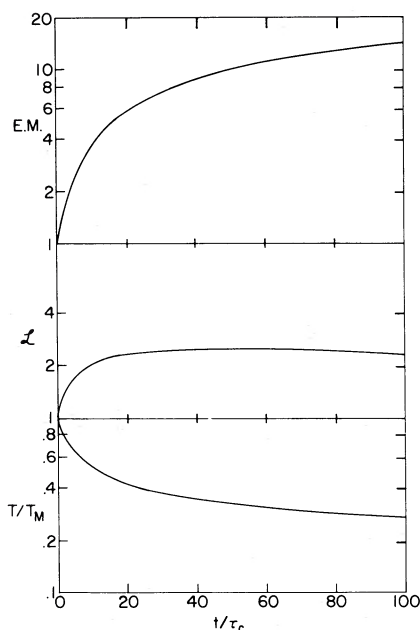


FIG. 3.—The evolution of the integrated soft X-ray emission $\mathcal{L}$ between 1 and 2 keV, the temperature T , and the emission measure (E.M.) for a loop of initial temperature $T_M = 2 \times 10^7$ K.

from 1 to 2 keV. We also plot the time dependence of the temperature and of the emission measure in Figure 3. Note that, even though the temperature is continuously decreasing, $\mathcal{L}$ increases initially because of the increasing emission measure and has a maximum at $t/\tau_c = 50$. For larger values of r_1 and r_2 —e.g., $r_1 = 2$ ($E_1 = 4$ keV) and $r_2 = 4$ ($E_2 = 8$ keV)—the exponential term in the emissivity (eq. [3.10]) dominates, and $\mathcal{L}$ decreases monotonically. We find that these results are insensitive to the compression factor Γ , because the X-ray emission comes mainly from the high-temperature material high in the loop.

In Figure 3, $\mathcal{L}$, the emission measure E.M., and T/T_M have been plotted for the maximum time interval over which we can expect our model to be accurate, $t \approx 10^2 \tau_c$. Of course, for particular values of the parameters T_M , p , and s_b (such as the ones given previously), the radiative losses may become important before $t = 10^2 \tau_c$. But, as long as conduction dominates initially, our result that evaporation may cause the maximum in $\mathcal{L}$ to occur during the cooling phase of the loop is still valid. The qualitative features of Figure 3 are in agreement with soft X-ray observations. However, we would not expect to find detailed agreement, since we have calculated the emission from only a single loop, whereas a flare is likely to consist of many loops at different stages in their evolution. In addition, we have neglected any heating which may be present during the decay phase of flares. However, our results do predict a rise in soft X-ray flux concurrent with a decrease in temperature, whereas in the static models the emission must decrease with the temperature.

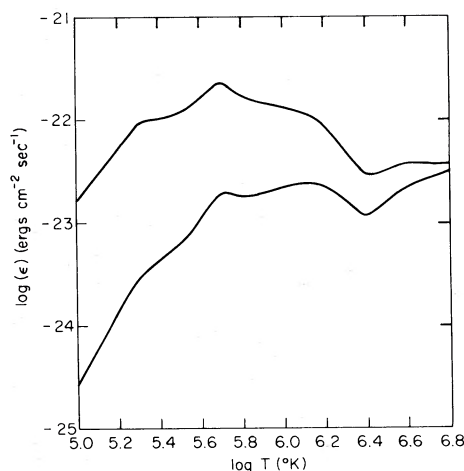


FIG. 4.—The radiative losses per unit logarithmic temperature interval of plasma in a loop as a function of temperature along the loop. The top line refers to an evaporative model, and the bottom line refers to a static model.

We believe that a strong observational test of our model may be provided by the spectrum of the radiation. From the form of the scale height, Figure 1, we note that an evaporative loop contains relatively more plasma at low temperatures ($T \lesssim 10^6$ K) than does a static one, which implies that the spectra of the emitted radiation (especially at EUV wavelengths) should be significantly different in the two cases. This is clearly illustrated in Figure 4, where we plot, for both models, the radiative losses $\epsilon(T)$ due to plasma at different temperatures along each leg of a loop; i.e.,

$$\epsilon(T) = Q(T)/2A, \quad (3.14)$$

where $Q(T)$ has been defined in equation (2.25). We use Cox and Tucker's (1969) values for $\Lambda(T)$, and H and n are in units of loop length s_b and of density at the top of the loop $n(s=0)$, respectively. The temperature at the top of the loop is assumed to be $T_M = 10^7$ K. We note that, in the static case, the highest-temperature plasma is responsible for most of the energy loss, whereas, in the evaporative loop, plasma at $\sim 5 \times 10^5$ K radiates most strongly.

In a later paper we intend to calculate the spectra for both models in detail—in particular, the emission in various EUV and X-ray lines—and to compare these calculations with recent *Skylab* observations.

One of us (P. A. S.) thanks Professors G. R. Burbidge and L. E. Peterson and their colleagues at the University of California, San Diego, for their kind hospitality in the spring of 1975, when this work was initiated. We also thank Mr. Keith Marzullo for his assistance with the computer calculations. This work was supported by the National Aeronautics and Space Administration under grant NGL 05-020-272 and the Office of Naval Research under contract N00014-75-C-0673.

REFERENCES

- Antiochos, S. K., and Sturrock, P. A. 1976, *Solar Phys.*, **49**, 359.
- Cox, D. P., and Tucker, W. H. 1969, *Ap. J.*, **157**, 1157.
- Craig, I. J. D., and Brown, J. C. 1976, *Astr. Ap.*, **49**, 239.
- Horan, D. M. 1971, *Solar Phys.*, **21**, 188.
- Hudson, H. S., and Ohki, K. 1972, *Solar Phys.*, **23**, 155.
- Jefferies, J. T., and Orrall, F. Q. 1965, *Ap. J.*, **141**, 519.
- Kahler, S. W., Meekins, J. F., and Kreplin, R. W. 1970, *Ap. J.*, **162**, 293.
- Neupert, W. M. 1968, *Ap. J. (Letters)*, **153**, L59.
- Phillips, K. J. H., and Neupert, W. M. 1973, *Solar Phys.*, **32**, 209.
- Raymond, J. C., Cox, D. P., and Smith, B. W. 1976, *Ap. J.*, **204**, 290.
- Rust, D. M., and Bar, V. 1973, *Solar Phys.*, **33**, 445.
- Spitzer, L. 1962, *Physics of Fully Ionized Gases* (New York: Interscience), chap. 5.
- Strauss, F. M., and Papagiannis, M. D. 1971, *Ap. J.*, **164**, 369.
- Sturrock, P. A. 1973, *Proc. Symp. on High Energy Phenomena on Sun*, Greenbelt, MD.
- Zaumen, W. T., and Acton, L. W. 1974, *Solar Phys.*, **30**, 139.

S. K. ANTIOCHOS and P. A. STURROCK: Institute for Plasma Research, Stanford University, Stanford, CA 94305